

Ku Kang
Chemical, Biological, Radiological and Nuclear (CBRN) Defense Research Institute
Seoul, Republic of Korea
bisu9082@gmail.com

19 September 2026

Dear Editors,

We are pleased to submit our manuscript, **“How much of hazard news is news? Decomposing open-source coverage of chemical accidents and CBRN threats for hazard surveillance,”** for consideration as a Research Article in *Safety Science*.

Safety and security functions increasingly read the volume of news coverage as a proxy for how often a hazard occurs, especially for chemical, biological, radiological and nuclear (CBRN) threats, for which no incident registry exists. The manuscript tests the assumption behind that practice: that an article about a hazard is evidence that something happened. Using 109,610 Korean news articles from 54 outlets over five years, for 30 queries in three blocks (industrial chemical accidents, CBRN security threats, conventional terrorism), we decompose news volume into five labelled components and show that only one article in three reports an event, one in ten for CBRN security hazards by model coding, and one in fifty a domestic CBRN event. Two of the authors independently coded a 400-article stratified subset (inter-coder $\kappa = 0.85$) and put the CBRN figure lower still, at one article in twenty (5.0%, 95% CI 2.5–7.9), and the domestic share at one in forty, because the model coders counted topically adjacent occurrences that the codebook excludes. We report which results that validation licenses and which remain on model labels. Query-level event shares, computed on the model labels, separate the CBRN block from both control blocks, which this design lacks the power to distinguish from each other; the separation survives conditioning on topical relevance, so it is not produced by the three query sets retrieving with different precision. We define a reference-bias metric (the share of articles citing a past anchor event), which reaches 89% for Novichok and 33% for hydrogen fluoride against 1–2% for matched controls, and show against the national chemical-accident registry that article counts for six substances are unrelated to incident counts. A lightweight text classifier recovers the event component, and the archive’s own incident taxonomy does not improve it. The human validation is also a result in its own right: two language models agreed with each other at $\kappa = 0.79$ while agreeing with human coders at 0.50, and at 0.24 in the CBRN block, which is a caution for the growing use of model-based annotation in this literature.

We believe the work fits *Safety Science* for three reasons. It sits squarely at the interface between a technology, an organisational practice and the people who act on it: the technology is a searchable news archive read by automated coders, the practice is sizing a hazard from what that archive returns, and the paper measures what that practice actually delivers. It gives a safety function an operational quantity, the event share with a confidence interval, for a proxy that is widely used and, so far as we can establish, never validated. And it benchmarks open-source coverage against the national chemical-accident registry, which to our knowledge has not been done: for six substances, article counts bear no detectable relation to recorded incidents. The industrial block is the part a process-safety reader will recognise most directly, and the CBRN and conventional-terrorism blocks are the controls that let us state that the pattern belongs to rare, feared and institutionally active hazards rather than to news in general. The paper closes with a procedure an organisation can run on its own hazard and archive in a day or two, and the codebook, anchor dictionary, labelled sample and analysis code are released so that it can be applied elsewhere.

The manuscript has not been published previously and is not under consideration by another journal. All authors have approved the submission. We declare no competing interests. Two large language models were used as independent coders under a written codebook, and language models assisted with drafting; both uses are stated in the manuscript, and the authors take full responsibility for the content. Article identifiers, the gold-set labels including both human coders' labels, the codebook, the registry extract and all analysis code are already public at <https://github.com/bisu9082/refbias>.

Thank you for considering our manuscript.

Sincerely,

Ku Kang, on behalf of all authors

Miae Kim, Janghyeon Cha, Myeongsik Shin, Gyungjun Choi, Ku Kang